

Course Outline - Summer 2026

MAT110: Differential Calculus and Coordinate Geometry

Faculty Name: Akram Hosen(RMH)

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Class Schedule: Please see from the classroom.

Counseling Hours: Please see from the classroom.

Recommended Books:

Text Books:

1. Calculus (10th Edition) by Howard Anton, Irl C. Bivens, Stephen Davis
2. A Text Book on Coordinate geometry and Vector Analysis by **Khosh Mohammad**
- 3.

Reference Books:

1. Calculus (9th Edition) by Thomas Finney
2. Calculus (9th edition) by James Stewart, Daniel K. Clegg, Saleem Watson, Lothar Redlin (2020, Cengage Learning)

Course Contents:

Calculus:

Functions, Limits, Continuity and Differentiability; Techniques of Differentiation; Successive Differentiations; Evaluation of Indeterminate forms by L'Hospitals rule; Leibnitz's Theorem; Increasing and Decreasing Function, Extrema (Maxima & Minima); Rolle's Theorem; Mean value Theorem; Taylor's Theorem and Maclaurin's Theorem (finite & infinite forms); Partial Derivatives; Chain rule of Partial Derivatives; Extrema (Maxima and Minima of Multivariable Functions); Optimization of Multivariable Functions; Taylor Expansion of Multivariable Functions; Lagrange Multipliers.

Co-ordinate Geometry:

Co-ordinate system, Transformation of co-ordinates, Jacobian in two-variables, Pair of straight lines, Conics (General equation of second degree, Circles, Parabola, Ellipse, Hyperbola).

Learning Outcomes:

This course is designed to provide science/ engineering students an intense foundational introduction to the fundamental concepts in mathematics. Upon completion of MAT110, a student will be able to:

- understand the concept of limit and its use to define derivative;
- explain the relationship between the derivative and linear approximation;
- develop his/her mathematical skills to solve extreme-value problems by implementing the knowledge of optimization;
- use rectangular and polar coordinates in 2D space;
- write down equations of lines, circles, and conics in 2D space and identify these curves from their equations;
- perform translations and rotations of the axes in 2D space;
- sketch and interpret mathematical graphs and functions and also find their domain and range;
- determine the role of mathematics as a logical, predictable system for expressing and relating quantities in analyzing and solving problems in the real World.

Course Evaluation:

We will follow the evaluation process of BRACU. The performance of the students will be evaluated throughout the semester by class tests/quizzes (average of best 3 out of 4), assignments, midterm and a comprehensive final exam. Students should attend all the quizzes/class tests and no make-up quiz exam will be arranged.

Marks Distribution:

Form of Assessment	Weight	Attendance	Marks
Attendance	5%	90% and above	5
Quiz (best three out of four)	20%	85% - 89%	4
Assignment (two and all count)	15%	80% - 84%	3
Midterm	20%	75% - 79%	2
Final Exam	40%	70% - 74%	1
Total	100%	Less than 70% (Unable to sit for Final Exam)	

Requirements/Regulations:

- Attendance at lectures (minimum 70% and above) are essential for successful completion of this course.
- Students are requested to follow the regulations of students conduct.

Grading Policy:

Mark Range	Grade	Grade Points
97 to $\leq$ 100	A+ (Exceptional)	4.0
90 to 96	A (Excellent)	4.0
85 to $<$ 90	A-	3.7
80 to $<$ 85	B+	3.3
75 to $<$ 80	B(good)	3.0
70 to $<$ 75	B-	2.7
65 to $<$ 70	C+	2.3
60 to $<$ 65	C (average)	2.0
57 to $<$ 60	C-	1.7
55 to $<$ 57	D+	1.3
52 to $<$ 55	D (poor)	1.0
50 to $<$ 52	D-	0.7
$<$ 50	F(Failure)	0.0

Lecture Plan:

Lecture No.	Topic(s)
1	Definition of function, Domain and range of a function with graph
2	Limits, Continuity of a function
3	Definition of derivative, Differentiability test
4	First principle of derivatives, Techniques of differentiation Quiz # 01
5	Successive differentiation; Indeterminate form (L' Hospital's rule)
6	Leibnitz's Theorem and solve related problems
7	Increasing and decreasing function, Extrema: Maxima and Minima
8	Rolle's Theorem, Mean-value Theorem Quiz # 02
9	Taylor's Theorem and Maclaurin's Theorem in finite and infinite forms
10	Review of the previous lectures
11	Midterm examination (tentative)
12	Partial derivatives; Chain Rules of Partial Derivatives
13	Extrema: Maxima and Minima of Multivariable Functions
14	Constrained optimization: Lagrange multipliers
15	Applications of Lagrange Multipliers (Optimization) Quiz # 03
16	Taylor expansion (Multivariable)
17	Co-ordinate system: Cartesian and polar, Transformation of Co-ordinates, Jacobian in two variables
18	Invariants theorem; Conics (General equation of second degree)
19	Pair of straight lines; Circles
20	Reducing the equation of Parabola to its standard form Quiz # 04
21	Reducing the equation of Ellipse and Hyperbola to their standard form
22	Review of the previous lectures

NOTE: Lecture plan may be changed, if necessary, according to the progress of the students.